

Exercise # 13.3

25-30

Find $\partial z / \partial x$ and $\partial z / \partial y$

25: $z = 4e^{x^2y^3}$

$$\frac{\partial z}{\partial x} = 4e^{x^2y^3}$$

$$\frac{\partial z}{\partial y} = 4e^{x^2y^3}$$

$$\frac{\partial z}{\partial x} = 4e^{x^2y^3} (y^3(2x))$$

$$\frac{\partial z}{\partial y} = 4e^{x^2y^3} (x^2(3y^2))$$

$$\frac{\partial z}{\partial x} = 8xy^3e^{x^2y^3}$$

$$\frac{\partial z}{\partial y} = 12x^2y^2e^{x^2y^3}$$

26: $z = \cos(x^5y^4)$

$$\frac{\partial z}{\partial x} = -\sin(x^5y^4)$$

$$\frac{\partial z}{\partial y} = -\sin(x^5y^4)$$

$$\frac{\partial z}{\partial x} = -\sin(\overbrace{x^5y^4}^{y^4(5x^4)})$$

$$\frac{\partial z}{\partial y} = -\sin(x^5y^4) \times (x^5(4y^3))$$

$$\frac{\partial z}{\partial x} = -5\sin(x^5y^4)x^4y^4$$

$$\frac{\partial z}{\partial y} = -\sin(x^5y^4)(x^54y^3)$$

$$\frac{\partial z}{\partial x} = -5\sin(x^5y^4)x^4y^4$$

$$\frac{\partial z}{\partial y} = -4x^5y^3\sin(x^5y^4)$$

$$\frac{\partial z}{\partial x} = -5x^4y^4\sin(x^5y^4)$$

$$27: z = x^3 \ln(1 + xy^{-3/5})$$

$$\because \ln(u) = \frac{1}{u} \cdot u'(x)$$

$$\frac{\partial z}{\partial x} = x^3 \cdot \frac{1}{1 + xy^{-3/5}} \times (1 + xy^{-3/5})$$

$$\frac{\partial z}{\partial x} = x^3 \cdot \frac{1}{1 + xy^{-3/5}} \times (0 + x^0 y^{-3/5})$$

$$\frac{\partial z}{\partial x} = x^3 \cdot \frac{1}{1 + xy^{-3/5}} \times (y^{-3/5})$$

$$\frac{\partial z}{\partial x} = \frac{x^3}{y^{3/5} (1 + xy^{-3/5})}$$

$$\frac{\partial z}{\partial x} = \frac{x^3}{y^{3/5} + x}$$

$$\frac{\partial z}{\partial y} = x^3 \frac{1}{1 + xy^{-3/5}} \times \left(\frac{-3}{5} y^{-3/5 - 1} x \right)$$

$$\frac{\partial z}{\partial y} = \frac{-3}{5} x^4 \cdot \frac{1}{y^{8/5} (1 + xy^{-3/5})}$$

$$\frac{\partial z}{\partial y} = \frac{-3}{5} \frac{x^4}{y^{8/5} + xy}$$

$$28: z = e^{xy} \sin 4y^2$$

$$\frac{\partial z}{\partial x} = y e^{xy} \sin 4y^2$$

$$\frac{\partial z}{\partial y} = 8y e^{xy} \cos 4y^2 + x e^{xy} \sin(4y^2)$$

$$29: z = \frac{xy}{x^2 + y^2}$$

Coefficient Rule

$$\frac{\partial z}{\partial x} = \frac{-y(x^2 - y^2)}{(x^2 + y^2)^2} \quad \frac{(x^2 + y^2)(y) - xy(2x)}{(x^2 + y^2)^2}$$

$$\frac{\partial z}{\partial x} = \frac{yx^2 + y^3 - 2x^2y}{(x^2 + y^2)^2} = \frac{-x^2y + y^3}{(x^2 + y^2)^2}$$

$$\frac{\partial z}{\partial x} = \frac{-y(x^2 - y^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial z}{\partial y} = \frac{(x^2 + y^2)(x) - (2y)xy}{(x^2 + y^2)^2}$$

$$30: z = \frac{x^2 y^3}{\sqrt{x+y}}$$

$$\frac{\partial z}{\partial y} = \frac{x^3 + xy^2 - 2xy^2}{(x^2 + y^2)^2}$$

$$\frac{\partial z}{\partial x} = \frac{\sqrt{x+y}(2xy^3) - \frac{x^2 y^3}{\sqrt{x+y}}}{(\sqrt{x+y})^{\frac{3}{2}}}$$

$$\frac{\partial z}{\partial y} = \frac{x^3 - xy^2}{(x^2 + y^2)^2}$$

$$\frac{\partial z}{\partial x} = \frac{2xy^3\sqrt{x+y} - \frac{x^2 y^3}{2\sqrt{x+y}}}{(x+y)^{\frac{3}{2}}}$$

$$\frac{\partial z}{\partial y} = \frac{x(x^2 - y^2)}{(x^2 + y^2)^2}$$

$$\frac{dz}{dx} = \frac{4xy^3(x+y) - x^2y^3}{2\sqrt{x+y}} \cdot \frac{1}{(x+y)}$$

$$\frac{dz}{dx} = \frac{4x^2y^3 + 4xy^4 - x^2y^3}{2(x+y)^{1/2}(x+y)^1}$$

$$\frac{dz}{dx} = \frac{3x^2y^3 + 4xy^4}{2(x+y)^{3/2}}$$

$$\frac{dz}{dy} = \frac{\sqrt{x+y}(x^2 \cdot 3y^2) - x^2y^3(x+y)^{1/2}}{(\sqrt{x+y})^2}$$

$$\frac{dz}{dy} = \frac{\sqrt{x+y} \cdot 3y^2x^2 - x^2y^3 \cdot \frac{1}{2}(x+y)^{-1/2}}{(x+y)}$$

$$\frac{dz}{dy} = \frac{x+y \cdot 3y^2x^2 - \frac{x^2y^3}{2\sqrt{x+y}}}{(x+y)}$$

$$\frac{dz}{dy} = \frac{6x^3y^2 + 6x^2y^3 - x^2y^3}{2(x+y)^{1/2}(x+y)^1}$$

$$\frac{dz}{dy} = \frac{6x^3y^2 + 5x^2y^3}{2(x+y)^{1/2+1}}$$

$$\frac{dz}{dy} = \frac{x^2y^2(6x+5y)}{2(x+y)^{3/2}}$$

31-36:

Find $f_x(x,y)$ and $f_y(x,y)$

$$31: f(x,y) = \sqrt{3x^5y - 7x^3y}$$

$$f_x(x,y) = \frac{1}{2} (3x^5y - 7x^3y)^{1/2-1} \cdot \frac{\partial}{\partial x} (3x^5y - 7x^3y)$$

$$f_x(x,y) = \frac{1}{2} (3x^5y - 7x^3y)^{-1/2} \cdot 3 \cdot 5x^4y - 7 \cdot 3x^2y$$

$$f_x(x,y) = \frac{1}{2} (3x^5y - 7x^3y)^{-1/2} \cdot (15x^4y - 21x^2y)$$

$$f_x(x,y) = \frac{1}{2} (3x^5y - 7x^3y)^{-1/2} \cdot 3x^2y(5x^2 - 7)$$

$$f_x(x,y) = \frac{3}{2} x^2y (5x^2 - 7) \cdot (3x^5y - 7x^3y)^{-1/2}$$

$$f_x(x,y) = \frac{3x^2y(5x^2 - 7)}{2\sqrt{3x^5y - 7x^3y}}$$

y-axis

$$f_y(x,y) = \sqrt{3x^5y - 7x^3y}$$

$$f_y(x,y) = \frac{1}{2} (3x^5y - 7x^3y)^{1/2-1} \cdot \frac{\partial}{\partial y} (3x^5y - 7x^3y)$$

$$f_y(x,y) = \frac{1}{2} (3x^5y - 7x^3y)^{-1/2} \cdot (3x^5 - 7x^3)$$

$$f_y(x,y) = \frac{1}{2} (3x^5y - 7x^3y)^{-1/2} \cdot (3x^5 - 7x^3)$$

$$f_y(x, y) = \frac{1}{2} (3x^5y - 7x^3y)^{-1/2} \cdot (3x^3 - 7x)$$

$$f_y(x, y) = \frac{1}{2} (3x^5y - 7x^3y)^{1/2} \cdot x^3 (3x^2 - 7)$$

$$f_y(x, y) = \frac{1}{2\sqrt{3x^5y - 7x^3y}} \cdot x^3 (3x^2 - 7)$$

$$f_y(x, y) = \frac{x^3(3x^2 - 7)}{2\sqrt{3x^5y - 7x^3y}}$$

32: $f(x, y) = \frac{x+y}{x-y}$

$$f(x, y) = \frac{\partial}{\partial x} \left[\frac{x+y}{x-y} \right]$$

$$f_x(x, y) = \frac{(x-y) \frac{\partial}{\partial x} (x+y) - (x+y) \frac{\partial}{\partial x} (x-y)}{(x-y)^2}$$

$$f_x(x, y) = \frac{(x-y) \cdot 1 - (x+y) \cdot 1}{(x-y)^2}$$

$$f_x(x, y) = \frac{x-y-x-y}{(x-y)^2}$$

$$f_x(x, y) = \frac{-2y}{(x-y)^2}$$

y-axis

$$f_y(x, y) = \frac{\partial}{\partial y} \left[\frac{x+y}{x-y} \right]$$

$$f_y(x, y) = \frac{(x-y) \frac{\partial}{\partial y} (x+y) - (x+y) \frac{\partial}{\partial y} (x-y)}{(x-y)^2}$$

$$f_y(x, y) = \frac{(x-y) \cdot 1 - (x+y) \cdot 1}{(x-y)^2}$$

$$f_y(x, y) = \frac{(x-y) + (x+y)}{(x-y)^2}$$

$$f_y(x, y) = \frac{x-y + x+y}{(x-y)^2}$$

$$f_y(x, y) = \frac{2x}{(x-y)^2}$$

33:

$$f(x, y) = y^{-3/2} \cdot \tan^{-1}\left(\frac{x}{y}\right)$$

$$f_x(x, y) = y^{-3/2} \cdot \tan^{-1}\left(\frac{x}{y}\right) \quad \because \tan^{-1} x = \frac{1}{1+x^2}$$

$$f_x(x, y) = y^{-3/2} \cdot \frac{\partial}{\partial x} \left[\tan^{-1}\left(\frac{x}{y}\right) \right]$$

$$f_x(x, y) = y^{-3/2} \cdot \frac{1}{1+\left(\frac{x}{y}\right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{x}{y} \right)$$

$$f_x(x, y) = y^{-3/2} \cdot \frac{1}{1+\frac{x^2}{y^2}} \cdot \frac{1}{y}$$

$$f_x(x, y) = \frac{y}{y^2 + x^2} \cdot y^{-3/2}$$

$$f_x(x, y) = \frac{y^{-3/2} \times y^1}{y^2 + x^2}$$

$$f_x(x, y) = \frac{y^{-3/2+1}}{y^2 + x^2}$$

$$f_x(x, y) = \frac{y^{-1/2}}{y^2 + x^2}$$

y-axis

$$f_y(x, y) = \frac{\partial}{\partial y} \left[y^{-3/2} \cdot \tan^{-1}\left(\frac{x}{y}\right) \right]$$

$$f_y(x, y) = y^{-3/2} \cdot \frac{\partial}{\partial y} \cdot \tan^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{x}{y}\right) \cdot \frac{\partial}{\partial y} (y^{-3/2})$$

$$f_y(x, y) = y^{-3/2} \cdot \frac{\partial}{\partial y} \cdot \tan^{-1}\left(\frac{x}{y}\right)$$

$$f_y(x, y) = y^{-3/2} \cdot \frac{1}{1 + \left(\frac{x}{y}\right)^2} \cdot \frac{\partial}{\partial y} \left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{x}{y}\right) \cdot \frac{-3}{2} y^{-5/2}$$

$$f_y(x, y) = y^{-3/2} \cdot \frac{1}{1 + \frac{x^2}{y^2}} \cdot \frac{\partial}{\partial y} (xy^{-1}) + \tan^{-1}\left(\frac{x}{y}\right) - \frac{3}{2} y^{-5/2}$$

$$f_y(x, y) = y^{-3/2} \cdot \frac{1}{\frac{y^2 + x^2}{y^2}} \cdot x \cdot -1y^{-1-1} + \tan^{-1}\left(\frac{x}{y}\right) - \frac{3}{2} y^{-5/2}$$

$$f_y(x, y) = \frac{y^{-3/2} \times y^2}{y^2 + x^2} \cdot \left(-\frac{x}{y^2}\right) + \tan^{-1}\left(\frac{x}{y}\right) \cdot \frac{-3}{2} y^{-5/2}$$

$$f_y(x, y) = \frac{y^{-3/2} \times y^2}{y^2 + x^2} \times \frac{-x}{y^2} + \tan^{-1}\left(\frac{x}{y}\right) \cdot \frac{-3}{2} y^{-5/2}$$

$$f_y(x, y) = \frac{x y^{-3/2}}{y^2 + x^2} + \tan^{-1}\left(\frac{x}{y}\right) \cdot \frac{-3}{2} y^{-5/2}$$

34: $F(x, y) = x^3 e^{-y} + y^3 \sec \sqrt{x}$

$$f_x(x, y) = \frac{\partial}{\partial x} [x^3 e^{-y} + y^3 \sec \sqrt{x}]$$

$$f_x(x, y) = \frac{\partial}{\partial x} (x^3 e^{-y}) + \frac{\partial}{\partial x} (y^3 \sec \sqrt{x})$$

$$f_x(x, y) = 3x^2 e^{-y} + y^3 \cdot \frac{\partial}{\partial x} (\sec \sqrt{x})$$

As Now, $\frac{\partial}{\partial x} (\sec x) = \sec x \tan x.$

$$f_x(x, y) = 3x^2 e^{-y} + y^3 \cdot \sec \sqrt{x} \cdot \tan x \cdot \frac{\partial}{\partial x} (x)^{1/2}$$

$$f_x(x, y) = 3x^2 e^{-y} + y^3 \cdot \sec \sqrt{x} \cdot \tan x \cdot \frac{1}{2} x^{-1/2}$$

$$f_x(x, y) = 3x^2 e^{-y} + \frac{y^3}{2\sqrt{x}} \cdot \sec \sqrt{x} \cdot \tan x$$

y-axis

$$f_y(x, y) = x^3 e^{-y} + y^3 \sec \sqrt{x}$$

$$f_y(x, y) = \frac{\partial}{\partial y} [x^3 e^{-y} + y^3 \sec \sqrt{x}]$$

$$f_y(x, y) = \frac{\partial}{\partial y} (x^3 e^{-y}) + \frac{\partial}{\partial y} (y^3 \sec \sqrt{x})$$

$$f_y(x, y) = \frac{\partial}{\partial y} (x^3 e^{-y}) + \frac{\partial}{\partial y} (y^3 \sec \sqrt{x})$$

$$f_y(x, y) = x^3 e^{-y} \cdot \frac{\partial}{\partial y} (-y) + 3y^2 \sec \sqrt{x}$$

$$f_y(x, y) = x^3 e^{-y} \cdot -1 + 3y^2 \sec \sqrt{x}$$

$$f_y(x, y) = x^3 e^{-y} + 3y^2 \sec \sqrt{x}$$

35: $f(x, y) = (y^2 \tan x)^{-4/3}$

$$f_x(x, y) = \frac{\partial}{\partial x} (y^2 \tan x)^{-4/3}$$

$$f_x(x, y) = \frac{-4}{3} (y^2 \tan x)^{-4/3-1} \cdot \frac{\partial}{\partial x} (\tan x) \cdot y^2$$

$$f_x(x, y) = \frac{-4}{3} (y^2 \tan x)^{-7/3} \cdot \sec^2 x$$

$$f_x(x, y) = \frac{-4}{3} y^2 \sec^2 x \cdot (y^2 \tan x)^{-7/3}$$

y-axis

$$f_y(x, y) = \frac{\partial}{\partial y} [(y^2 + \tan x)^{-4/3}]$$

$$f_y(x, y) = -\frac{4}{3} (y^2 + \tan x)^{-4/3-1} \frac{d}{dy} (y^2) \cdot \tan x$$

$$f_y(x, y) = -\frac{4}{3} (y^2 + \tan x)^{-7/3} (2y) \cdot \tan x$$

$$f_y(x, y) = -\frac{4}{3} (y^2 + \tan x)^{-7/3} \cdot (2y) \cdot \tan x$$

$$f_y(x, y) = -\frac{8}{3} y \tan x (y^2 + \tan x)^{-7/3}$$

36:

$$f(x, y) = \cosh(\sqrt{x}) \sinh^2(xy^2)$$

$$f_x(x, y) = \frac{\partial}{\partial x} [\cosh(\sqrt{x}) \sinh^2(xy^2)]$$

using product Rule

$$f_x(x, y) = \cosh \sqrt{x} \cdot \frac{\partial}{\partial x} (\sinh^2(xy^2)) + \sinh^2(xy^2) \cdot \frac{\partial}{\partial x} (\cosh \sqrt{x})$$

$$\frac{\partial}{\partial x} (\cosh \sqrt{x})$$

$$f_x(x, y) = \cosh \sqrt{x} \cdot 2 \sinh(xy^2) \cdot \frac{\partial}{\partial x} (\sinh(xy^2))$$

$$f_x(x, y) = \cosh \sqrt{x} \cdot 2 \sinh(xy^2) \cdot \frac{\partial}{\partial x} (\cosh \sqrt{x})$$

$$+ \sinh^2(xy^2) \cdot \frac{\partial}{\partial x} (\sinh(xy^2))$$

$$f_x(x, y) = + \sinh^2(xy^2) \cdot \sinh \sqrt{x} \cdot \frac{d}{dx} (x)^{1/2}$$

$$f_x(x, y) = \cosh \sqrt{x} \cdot 2 \sinh(xy^2) \cdot \cosh(xy^2) \cdot (y^2) \\ + \sinh^2(xy^2) \cdot \sinh \sqrt{x} \cdot \frac{1}{2} x^{-1/2} \cdot 1$$

$$f_x(x, y) = 2y^2 \cosh \sqrt{x} \cdot \sinh(xy^2) \cdot \cosh(xy^2) \\ + \frac{1}{2} x^{-1/2} \cdot \sinh \sqrt{x} \cdot \sinh^2(xy^2)$$

y-axis

$$f_y(x, y) = \cosh(\sqrt{x}) \sinh^2(xy^2)$$

$$f_y(x, y) = \frac{\partial}{\partial y} [\cosh(\sqrt{x}) \cdot \sinh^2(xy^2)]$$

$$f_y(x, y) = \cosh \sqrt{x} \cdot \frac{\partial}{\partial y} (\sinh^2(xy^2))$$

$$f_y(x, y) = \cosh \sqrt{x} \cdot 2 \sinh(xy^2) \cdot \cosh(xy^2)$$

$$f_y(x, y) = 4xy \cosh \sqrt{x} \cdot \sinh(xy^2) \cdot \cosh(xy^2)$$

Ans

↓ Mix step

$$f_y(x, y) = \cosh \sqrt{x} \cdot 2 \sinh(xy^2) \cdot \frac{\partial}{\partial y} (\sinh(xy^2) \cdot 2xy)$$

Evaluate the indicated derivatives

37:

$$f(x, y) = 9 - x^2 - 7y^3 \quad \begin{matrix} f_x(3, 1) \\ f_y(3, 1) \end{matrix}$$

$$f_x(x, y) = -2x$$

$$f_x(3, 1) = -2(3)$$

$$f_x(3, 1) = -6$$

$$f_y(x, y) = -21y^2$$

$$f_y(3, 1) = -21(1)^2$$

$$f_y(3, 1) = -21$$

$$38: f(x, y) = x^2 y e^{xy} \quad \frac{\partial f}{\partial x}(1, 1)$$

$$\frac{\partial f}{\partial x} = \frac{y \cdot (x^2) \cdot e^{xy}(y)}{y^2 x e^{xy}} + \frac{\partial f}{\partial f}(1, 1)$$

$$\frac{\partial f}{\partial x} = x^2 y^2 e^{xy} + 2xy e^{xy}$$

$$\frac{\partial f}{\partial x}(1, 1) = (1)^2 e^{(1)} + 2(1) e^{(1)}$$

$$= e + 2e = 3e$$

$$\frac{\partial f}{\partial y} = x^2(y) e^{xy}(x) + x^2(1) e^{xy}$$

$$\frac{\partial f}{\partial y} = (1) \cdot e^{(1)} + e^{(1)}$$

$$= e + e = 2e$$

$$39: z = \sqrt{x^2 + 4y^2}$$

$$\frac{\partial z}{\partial x} (1, 2)$$

$$\frac{\partial z}{\partial x} = \left(\sqrt{x^2 + 4y^2} \right)^2$$

$$\frac{\partial z}{\partial y} (1, 2)$$

$$\frac{\partial z}{\partial x} = \frac{1}{2} (x^2 + 4y^2)^{-1/2} (2x)$$

$$\frac{\partial z}{\partial x} = x (x^2 + 4y^2)^{-1/2}$$

$$\frac{\partial z}{\partial x} = (1)(1)^{-1/2} = 1/\sqrt{17}$$

$$\frac{\partial z}{\partial y} = 4y (x^2 + 4y^2)^{-1/2}$$

$$\frac{\partial z}{\partial y} (1, 2) = 8/\sqrt{17}$$

$$40: w = x^2 \cos xy; \frac{\partial w}{\partial x} \left(\frac{1}{2}, \pi \right)$$

$$\frac{\partial w}{\partial y} \left(\frac{1}{2}, \pi \right)$$

$$x^2(-\sin xy)(y) + 2x \cos xy$$

$$\frac{\partial w}{\partial x \left(\frac{1}{2}, \pi \right)} = -x^2 y \sin xy + 2x \cos xy$$

$$\frac{\partial w}{\partial y} = -x^3 \sin xy$$

$$\frac{\partial w}{\partial y \left(\frac{1}{2}, \pi \right)} = -1/8$$

$$41: \text{Let } f(x, y, z) = x^2 y^4 z^3 + xy + z^2 + 1, \text{ Find}$$

$$(a) f_x(x, y, z) = 2xy^4z^3 + y$$

$$(b) f_y(x, y, z) = 4x^2y^3z^3 + x$$

$$(c) f_z(x, y, z) = 3x^2y^4z^2 + 2z$$

$$(d) f_x(1, y, z) = 2y^4z^3 + y$$

$$(e) f_y(1, 2, z) = 32z^3 + 1$$

$$(f) f_z(1, 2, 3) = 438$$

42) Let $w = x^2 y \cos z$. find

a) $\partial w / \partial x (x, y, z) = 2xy \cos z$

b) $\partial w / \partial y (x, y, z) = x^2 \cos z$

c) $\partial w / \partial z (x, y, z) = -x^2 y \sin z$

d) $\partial w / \partial x (2, y, z) = 4y \cos z$

e) $\partial w / \partial y (2, 1, z) = 4 \cos z$

f) $\partial w / \partial z (2, 1, 0) = 0$

43) Find f_x, f_y and f_z

fin $f(x, y, z) = z \ln(x^2 y \cos z)$

$$f_x = 2z/x = \frac{1}{x^2 y \cos z} \cdot y \cos z (2x)$$

$$f_y = z/y = z \frac{1}{x^2 y \cos z} \cdot x^2 \cos z$$

$$f_z = \ln(x^2 y \cos z) - z \tan z$$

$$(1) \ln(x^2 y \cos z) + 2 \frac{1}{x^2 y \cos z} - x^2 y \cos z$$

44) $f(x, y, z) = y^{-3/2} \sec \frac{xz}{y}$

$$y^{-3/2-1} \cdot 2$$

$$f_x = y^{-5/2} z \sec(xz/y) \tan(xz/y)$$

$$-3/2 y^{-3/2} - 1$$

$$f_y = xy^{-7/2} \sec(xz/y) \tan(xz/y) - (3/2)y^{-5/2}$$

$$f_z = xy^{-5/2} \sec(xz/y) \tan(xz/y) \sec(xz/y)$$

$$45. f(x, y, z) = \tan^{-1}\left(\frac{1}{xy^2z^3}\right)$$

$$f_x = -y^2z^3 / (1 + x^2y^4z^6)$$

$$= \frac{1}{1 + \left(\frac{1}{x^2y^2z^3}\right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{1}{xy^2z^3} \right)$$

$$= \frac{x^2 \cdot y^4 \cdot z^6}{x^2y^4z^6 + 1} \left(-\frac{1}{x^2y^2z^3} \right)$$

$$f_x = -y^2z^3 / x^2y^4z^6 + 1$$

$$f_y = -2xy^2z^3 / (1 + x^2y^4z^6)$$

$$f_z = -3xy^2z^2 / (1 + x^2y^4z^6)$$

$$46. f(x, y, z) = \cosh(\sqrt{z}) \sinh^2(x^2yz)$$

$$f_x(x, y, z) = \frac{\partial}{\partial x} (f(x, y, z))$$

$$= \frac{\partial}{\partial x} [\cosh(\sqrt{z}) \sinh^2(x^2yz)]$$

$$= \cosh(\sqrt{z}) 2 \sinh(x^2yz) \cosh(x^2yz) yz \cdot 2x$$

$$= 2xyz \sinh 2(x^2yz) \cosh(\sqrt{z})$$

$$f_x = 2xyz \sinh 2(x^2yz) \cosh(\sqrt{z})$$

47-50 Find $\partial w / \partial x$, $\partial w / \partial y$ and $\frac{\partial w}{\partial z}$

47: $w = ye^z \sin xz$ — (1)

Differentiate on the both sides of the eq (1) with respect to x by treating y and z constant.

$$\begin{aligned} \frac{\partial w}{\partial x} &= \frac{\partial}{\partial x} (ye^z \sin(xz)) \\ &= ye^z \frac{\partial}{\partial x} (\sin(xz)) \end{aligned}$$

$\therefore$ chain rule $\frac{\partial}{\partial x} (f(g(x))) = f'(g(x))g'(x)$

$$\begin{aligned} \text{and } \frac{\partial}{\partial x} (\sin(x)) &= \cos x \\ &= ye^z (\cos(xz)(z) \frac{\partial}{\partial x} (x)) \\ &= ye^z \cos(xz)(1) \end{aligned}$$

$$\frac{\partial w}{\partial x} = yze^z \cos(xz)$$

48: $w = \frac{x^2 - y^2}{y^2 + z^2}$

$$\begin{aligned} \frac{\partial w}{\partial x} &= \frac{\partial}{\partial x} \left[\frac{x^2 - y^2}{y^2 + z^2} \right] \\ &= \frac{(y^2 + z^2) 2x - (x^2 - y^2)(0)}{(y^2 + z^2)^2} \end{aligned}$$

$$\frac{\partial w}{\partial y} = e^z \sin xz$$

$$\frac{\partial w}{\partial y} = ye^z (\sin xz + x \cos xz)$$

$$\frac{\partial w}{\partial x} = \frac{2x}{(y^2+z^2)}, \quad \frac{\partial w}{\partial y} = \frac{-2y(x^2+z^2)}{(y^2+z^2)^2}$$

$$\frac{\partial w}{\partial z} = \frac{2z(y^2-x^2)}{(y^2+z^2)^2}$$

49: $w = \sqrt{x^2+y^2+z^2}$

1; 12; 04

1; 41; 37

$$w = (x^2+y^2+z^2)^{1/2}$$

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x} [(x^2+y^2+z^2)^{1/2}]$$

$$= \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2x$$

50: $w = y^3 e^{2x+3z}$

$$\frac{\partial w}{\partial x} = \frac{x}{\sqrt{x^2+y^2+z^2}}$$

$$\frac{\partial w}{\partial x} = 2y^3 e^{2x+3z}$$

$$\frac{\partial w}{\partial y} = \frac{y}{\sqrt{x^2+y^2+z^2}}$$

$$\frac{\partial w}{\partial y} = 3y^2 e^{2x+3z}$$

$$\frac{\partial w}{\partial z} = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$\frac{\partial w}{\partial z} = 3y^3 e^{2x+3z}$$

101-104: Show that the Function satisfies Laplace equation.

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

a) $z = x^2 - y^2 + 2xy$

partial derivative w.r.t , ~~x~~ - axis

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (x^2 - y^2 + 2xy)$$

$$\frac{\partial z}{\partial x} = 2x - 0 - 2(1)y$$

$$\frac{\partial z}{\partial x} = 2x - 2y$$

again - x-axis Partial-D

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} (2x - 2y)$$

$$\frac{\partial^2 z}{\partial x^2} = 2$$

$$z = x^2 - y^2 + 2xy$$

partial derivative w.r.t , y-axis

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} (x^2 - y^2 + 2xy)$$

$$\frac{\partial z}{\partial y} = 0 - 2y + 2x(1)$$

$$\frac{\partial z}{\partial y} = -2y + 2x$$

Again partial derivative

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} (-2y + 2x)$$

$$\frac{\partial^2 z}{\partial y^2} = -2(1) + 0$$

As we know

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

$$2 + (-2) = 0$$

$$2 - 2 = 0$$

$$0 = 0$$

102 : Show that the function satisfy the heat equation.

$$\frac{\partial z}{\partial t} = c^2 \frac{\partial^2 z}{\partial x^2} \quad (c > 0) \text{ constant}$$

a) $z = e^t \cdot \sin(x/c)$

partial derivative w-r-t, t-axis

$$\frac{\partial z}{\partial t} = \frac{\partial}{\partial t} (e^t \cdot \sin(\frac{x}{c}))$$

$$\frac{\partial z}{\partial t} = e^{-t} \cdot -1 \cdot \sin\left(\frac{x}{c}\right)$$

$$\frac{\partial z}{\partial t} = -e^{-t} \cdot \sin\left(\frac{x}{c}\right)$$

partial derivative w.r.t, x-axis

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} \left(e^{-t} \cdot \sin\left(\frac{x}{c}\right) \right)$$

$$\frac{\partial z}{\partial x} = e^{-t} \cdot \cos\left(\frac{x}{c}\right) \cdot \frac{\partial}{\partial x} \left(\frac{x}{c} \right)$$

$$\frac{\partial z}{\partial x} = e^{-t} \cdot \cos\left(\frac{x}{c}\right) \cdot \frac{1}{c}$$

$$\frac{\partial z}{\partial x} = \frac{1}{c} e^{-t} \cdot \cos\left(\frac{x}{c}\right)$$

Again partial derivative

$$\frac{\partial^2 z}{\partial x^2} = \frac{1}{c} \frac{\partial}{\partial x} \left(e^{-t} \cdot \cos\left(\frac{x}{c}\right) \right)$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{1}{c} \cdot e^{-t} \cdot -\sin\left(\frac{x}{c}\right) \cdot \frac{1}{c}$$

$$\frac{\partial^2 z}{\partial x^2} = \left(\frac{1}{c^2} \right) e^{-t} \cdot -\sin\left(\frac{x}{c}\right)$$

$$\frac{\partial^2 z}{\partial x^2} = -\left(\frac{1}{c^2} \right) e^{-t} \cdot \sin\left(\frac{x}{c}\right)$$

As we know that

$$\frac{\partial z}{\partial t} = c^2 \frac{\partial^2 z}{\partial x^2}$$

$$\frac{\partial z}{\partial t} - c^2 \frac{\partial^2 z}{\partial x^2} = 0$$

$$e^{-t} \sin\left(\frac{x}{c}\right) - c^2 \left(-\frac{1}{c^2}\right) e^{-t} \sin\left(\frac{x}{c}\right) = 0$$

$$-\cancel{e^{-t} \sin\left(\frac{x}{c}\right)} + \cancel{e^{-t} \sin\left(\frac{x}{c}\right)} = 0$$

$$0 = 0$$

104: show that $u(x,y)$ and $v(x,y)$ satisfy the Cauchy Riemann equation.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

As we know that

partial derivative w.r.t x -axis

$$u = x^2 - y^2, \quad v = 2xy$$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (x^2 - y^2)$$

$$\frac{\partial u}{\partial x} = 2x - 0$$

$$\frac{\partial u}{\partial x} = 2x \longrightarrow \text{Q.E.D.}$$

partial derivative w-r-t y-axis
in $V = 2xy$

$$\frac{\partial V}{\partial y} = \frac{\partial}{\partial y}(xy)$$

$$\frac{\partial V}{\partial y} = 2x \rightarrow (2)$$

Compare eq (1) and (2)

$$u = x^2 - y^2$$

partial derivative w-r-t y-axis

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(x^2 - y^2)$$

$$\frac{\partial u}{\partial y} = 0 - 2y$$

$$\frac{\partial u}{\partial y} = -2y \rightarrow (3)$$

$$V = 2xy$$

partial derivative w-r-t x-axis

$$\frac{\partial V}{\partial x} = 2 \cdot \frac{\partial}{\partial x}(xy)$$

$$\frac{\partial V}{\partial x} = 2 \cdot y$$

$$\frac{\partial V}{\partial x} = 2y \Rightarrow -\frac{\partial V}{\partial x} = -2y$$

$$-\frac{\partial V}{\partial x} = -2y \rightarrow (4)$$

Compare eq (3) and (4)

$$-2y = -2y$$